

Free cash flow level, margin, and yield

Directed factor-mimicking portfolio catalogue

Abstract

This in-sample catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by directed raw Sharpe, highest first. Direction is the sign that makes comparison-window Sharpe non-negative.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes direction and Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only.

Construction Configuration

Setting	Effective value
Factors	3
Universe	shared security screen; refreshed monthly
Formation filters	price ≥ 1.0 ; 21d dollar volume $\geq$ none
Market-cap screen	top 37.5% monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2007-03-01 to 2026-05-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ

Returns PCA

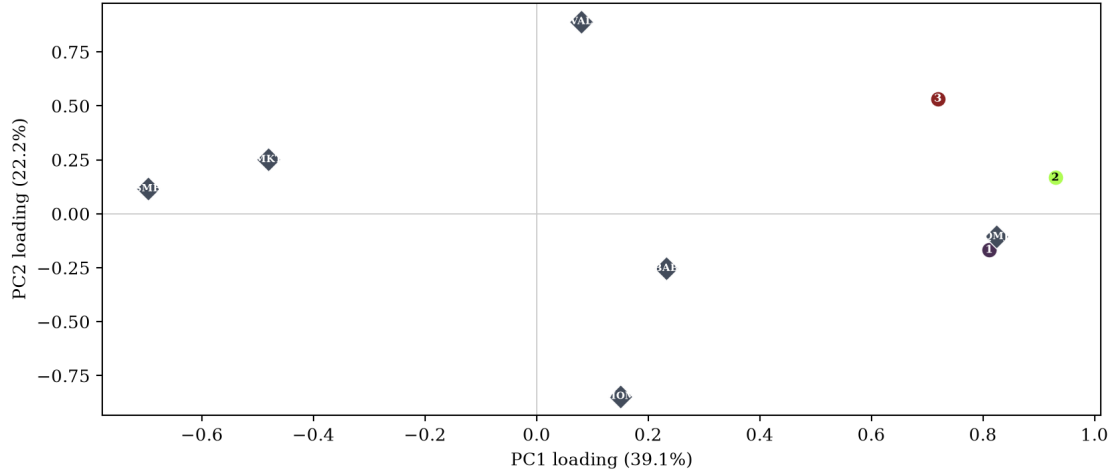


Figure: catalogue factors are numbered by directed-Sharpe index; attribution controls are named; $N=231$ common months.

Index

Idx	Factor	Dir.	N
1	FCFMargin	+1	231
2	FCF	+1	231
3	FCFYield	+1	231

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
FCFMargin	5.62%	5.90%	0.95	3.51	-15.83%	0.416	3.40%	5.90%
FCF	4.93%	6.91%	0.71	2.55	-21.79%	0.644	2.90%	6.91%
FCFYield	5.10%	8.02%	0.64	2.32	-28.91%	0.567	2.63%	8.02%

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
FCFMargin	0.05 (2.18)	-0.21 (-4.97)	-0.08 (-1.32)	0.02 (0.35)	-0.02 (-0.54)	0.28 (5.83)	3.85% (2.89)	0.423
FCF	0.05 (2.17)	-0.37 (-8.83)	0.19 (3.53)	0.06 (2.15)	-0.01 (-0.24)	0.34 (7.08)	3.27% (2.98)	0.648
FCFYield	0.12 (4.91)	-0.11 (-2.20)	0.33 (5.02)	-0.00 (-0.07)	-0.01 (-0.43)	0.49 (9.20)	2.81% (2.01)	0.568

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
FCFMargin	0.61%	0.14%	0.16%	0.06%	-0.05%	0.86%	3.85%	5.62%
FCF	0.57%	0.24%	-0.37%	0.22%	-0.02%	1.02%	3.27%	4.93%
FCFYield	1.45%	0.08%	-0.67%	-0.01%	-0.05%	1.49%	2.81%	5.10%

1. FCFMargin

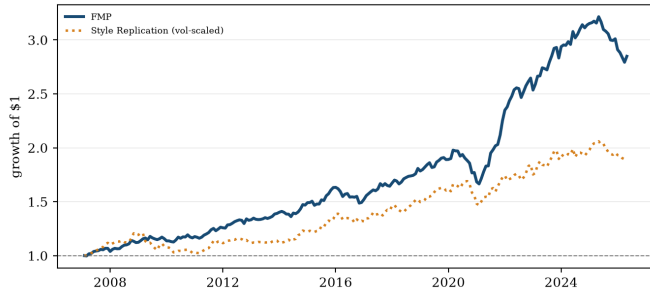


Figure: Common-window cumulative growth of \$1.

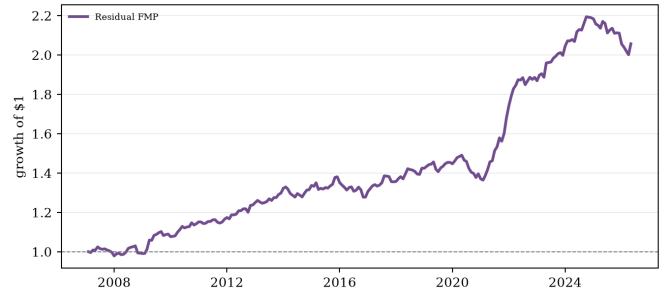


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

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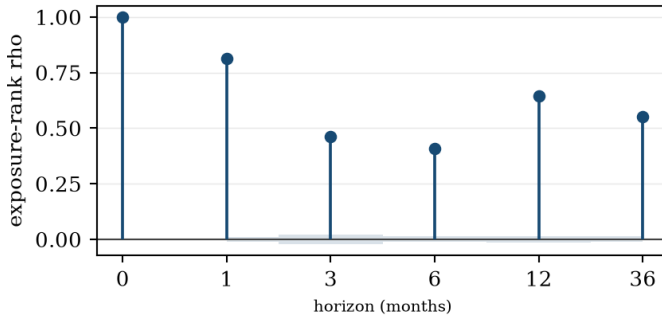


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

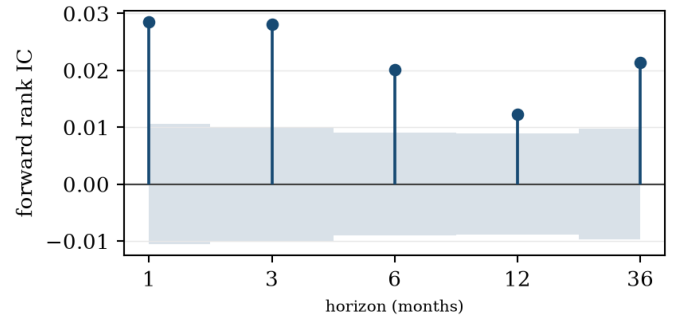


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	231
Signals	fcf_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 231
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 12
Stocks, min / median / max	495 / 718 / 1294
Available-history Sharpe	1.03

2. FCF

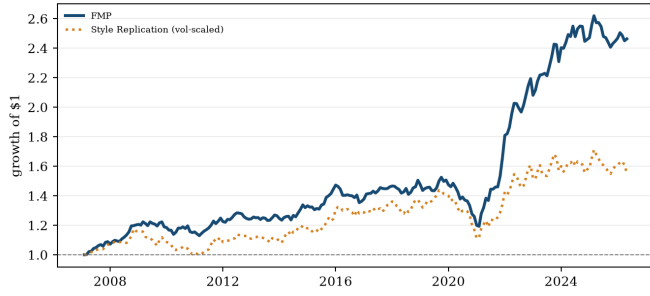


Figure: Common-window cumulative growth of \$1.

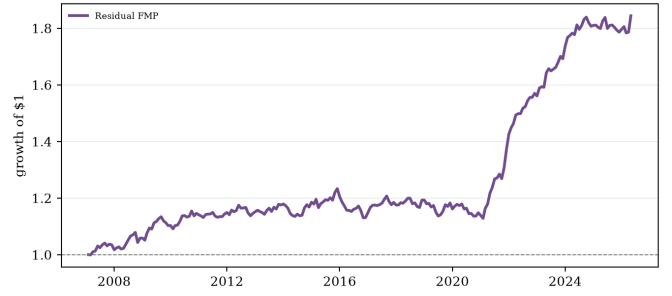


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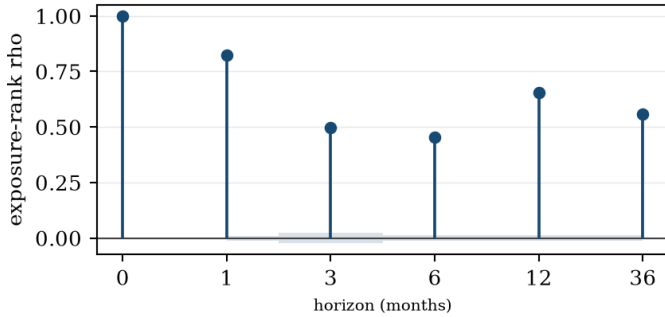


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

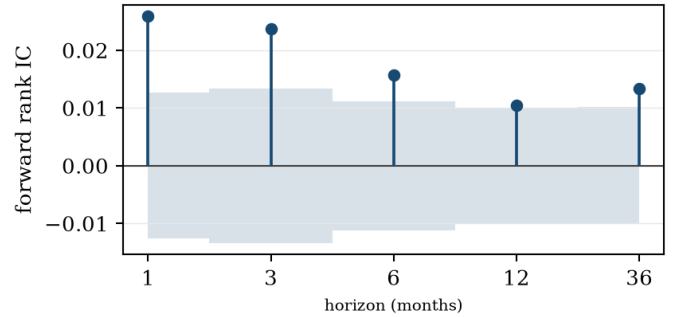


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	231
Signals	fcf(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	243 / 231
Available window	2006-05-01 to 2026-07-01
Dropped: low breadth / common	54 / 12
Stocks, min / median / max	495 / 720 / 1320
Available-history Sharpe	0.74

3. FCFYield

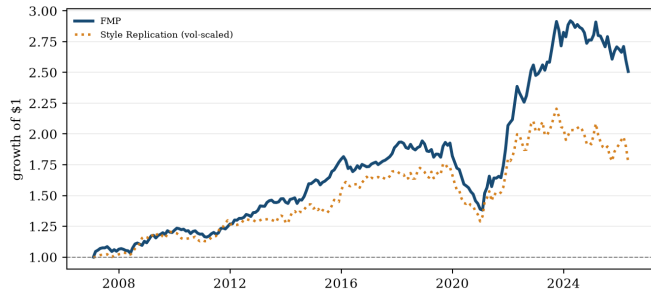


Figure: Common-window cumulative growth of \$1.

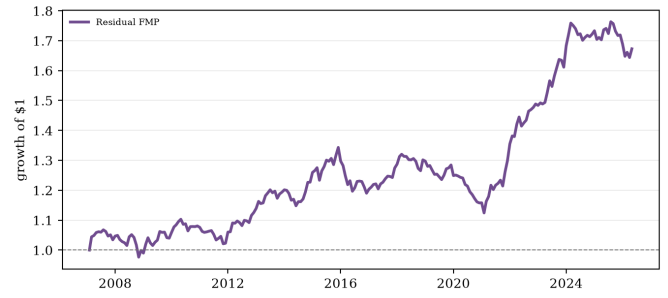


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

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Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
FCFYield	0.12 (4.91)	-0.11 (-2.20)	0.33 (5.02)	-0.00 (-0.07)	-0.01 (-0.43)	0.49 (9.20)	2.81% (2.01)	0.568

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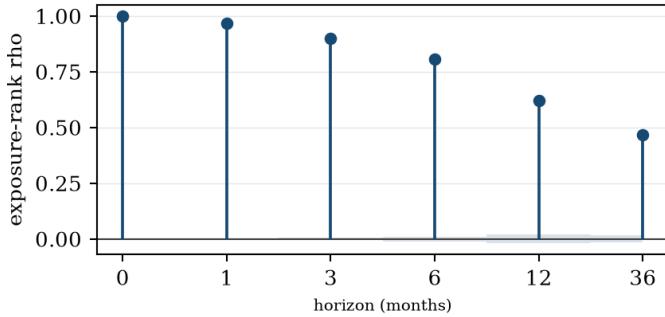


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

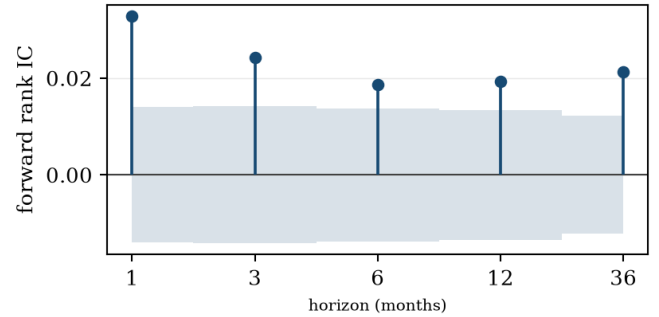


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	231
Signals	fcf_yield(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	233 / 231
Available window	2007-03-01 to 2026-07-01
Dropped: low breadth / common	15 / 2
Stocks, min / median / max	37 / 666 / 1260
Available-history Sharpe	0.66

Reproducibility

Run ID: 20260829T041533Z_free-cash-flow-level-margin-and-yield_743bfe3900.

Gold snapshot: version=20260826T131322Z.

Configuration hash: 743bfe3900.

Symbols requested / loaded: all / 5857.

Seed: 0. Comparison months: 231, 2007-03-01 to 2026-05-01.

Command: C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/
factor_model_runner/run.py--configsrc/experiments/factor_model_runner/fcf_set.py--n0--seed0.

Direction is selected once from raw comparison-window Sharpe and then applied before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.